

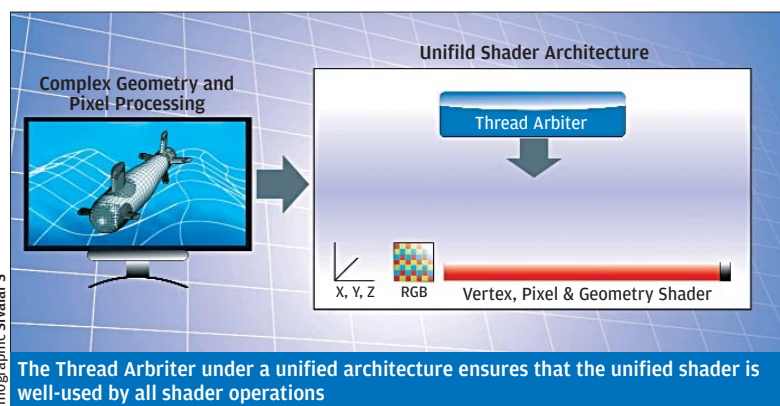
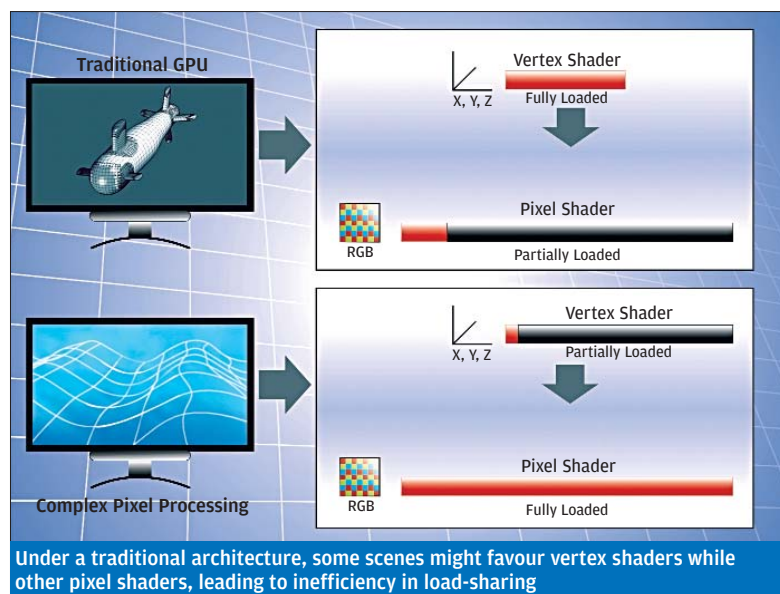
shares another similarity with the Xbox 360: like the 360, a GPU under DX10 can stream out data to the card's memory and then call it back in for further use within the GPU, without having to write to an external memory or disk resource. This greatly speeds things up and will allow for some interesting effects, such as realistic shadows.

Free The Processor

DX10 adds the geometry shader to the API lexicon. Simply put, a geometry shader deals with, well, geometry: every 3D object is made up of triangles, and a geometry shader can act on a bunch of vertices simultaneously. Thus it deals with shapes. To better understand its function, let's consider a practical example: a smooth curve is represented as a series of jointed straight lines within a GPU—the more the number of constituent lines, the smoother the curve. This is known as tessellation, and is a very CPU-intensive task (how many times have you seen a car in a game with a polygonal wheel instead of a smooth circular one?). The geometry shader can perform tessellation within the GPU itself, and since the GPU is much more powerful than a CPU when it comes to floating-point calculations, it can do so faster. Plus, the GPU need not send data outside. The geometry shader can similarly create new triangles around a point, or



By offloading tasks from the CPU, the GPU will allow the processor to perform complex physics calculations as seen here in *Cellfactor Revolution*



extrude the sides of a triangle to turn it into a volume, and so on.

The important thing to take from this is that a geometry shader further allows a GPU to offload work from the CPU. By reducing the amount of data that a GPU needs to outsource, two things happen—work is done quicker, and the CPU is free to do more. The graphics card can have a bigger hand in some of the tasks that were done by the CPU—things like rigid-body collision detection, calculating obscured geometry, audio acceleration, detailed shadows, and so on. The CPU, meanwhile, can work on other elements that add up to make a game more believable—such as advanced AI calculations and advanced physics simulations.

Sundry Bits

All this discussion was technically limited to the Direct3D element of DirectX 10. What about the other bits that constitutes DirectX? Can we expect major changes there? In a word, no!

One interesting feature does deserve mention though. With the next DirectInput, Microsoft has decided to blur the lines between its console offerings and the PC: you can take any Xbox 360 input device—be it a gamepad or a racing wheel—and use it under Windows. This is currently possible under Windows XP as well, and is thus not a feature exclusive to DX10. It is however, an evolution of the DirectInput constituent. DirectSound does not see any major changes.

Take It Outside

Traditionally, a graphics API has played in the kernel space of the OS. The result is the all-too-familiar system crashes following an errant driver install. With Windows Vista, Microsoft is taking security to heart. Major parts of the API will now be part of the user space of the OS. This means that the bits and bytes that make up graphics will not play in the same space that is meant for the essential parts of the OS. Not only